

AREA – Complete Formula Sheet with Full Forms

Q1. Right Triangle Area

Pythagoras Theorem

$$(\text{Hypotenuse})^2 = (\text{Perpendicular})^2 + (\text{Base})^2$$

$$(\text{Perpendicular})^2 = (\text{Hypotenuse})^2 - (\text{Base})^2$$

Area of Triangle

$$\text{Area} = (1/2) \times \text{Base} \times \text{Perpendicular}$$

where:

- Hypotenuse = longest side of right triangle
- Perpendicular = height of triangle
- Base = bottom side of triangle

Q2. Triangle Area with Ratio

Area of triangle:

$$\text{Area} = (1/2) \times \text{Base} \times \text{Height}$$

If Base : Height = 3 : 4

Let:

- Base = $3x$
- Height = $4x$

Then:

$$150 = (1/2) \times 3x \times 4x$$

Q3. Heron's Formula

$$\text{Semi-perimeter} = (\text{Side1} + \text{Side2} + \text{Side3}) / 2$$

$$\text{Area} = \sqrt{s(s - a)(s - b)(s - c)}$$

where:

- s = semi-perimeter
- a, b, c = sides of triangle

Q4. Equilateral Triangle

$$\text{Area} = (\sqrt{3} / 4) \times (\text{Side})^2$$

where:

- Side = length of each side

Q5. Isosceles Triangle

$$\text{Perimeter} = (2 \times \text{Equal Side}) + \text{Base}$$

$$\text{Equal Side} : \text{Base} = 5 : 4$$

$$\text{Height} = \sqrt{[(\text{Equal Side})^2 - (\text{Base}^2 / 4)]}$$

$$\text{Area} = (1/2) \times \text{Base} \times \text{Height}$$

Q6. Isosceles Triangle Perimeter

$$\text{Perimeter} = (2 \times \text{Equal Side}) + \text{Base}$$

Q7. Square and Equilateral Triangle

Square

$$\text{Diagonal} = \text{Side} \times \sqrt{2}$$

$$\text{Side} = \text{Diagonal} / \sqrt{2}$$

$$\text{Perimeter} = 4 \times \text{Side}$$

Equilateral Triangle

$$\text{Perimeter} = 3 \times \text{Side}$$

$$\text{Area} = (\sqrt{3} / 4) \times (\text{Side})^2$$

Q8. Diagonal of Rectangle

$$\text{Diagonal} = \sqrt{[(\text{Length})^2 + (\text{Breadth})^2]}$$

where:

- Length = longer side
- Breadth = shorter side

Q9. Rectangle and Fencing

$$\text{Perimeter} = 2 \times (\text{Length} + \text{Breadth})$$

$$\text{Cost} = \text{Perimeter} \times \text{Rate}$$

$$\text{Length} = \text{Breadth} + 20$$

Q10. Rectangle Ratio

$$\text{Perimeter} = 2 \times (\text{Length} + \text{Breadth})$$

$$\text{Length} : \text{Breadth} = 5 : 3$$

Let:

- Length = $5x$
- Breadth = $3x$

$$\text{Difference} = \text{Length} - \text{Breadth}$$

Q11. Rectangle Area Using Diagonal

$$(\text{Diagonal})^2 = (\text{Length})^2 + (\text{Breadth})^2$$

$$\text{Area} = \text{Length} \times \text{Breadth}$$

Q12. Rectangle with Changed Length

$$\text{Perimeter} = 2 \times (\text{Length} + \text{Breadth})$$

$$\text{Area} = \text{Length} \times \text{Breadth}$$

$$\text{New Length} = 4 \times \text{Length}$$

New Perimeter:

$$2 \times (4 \times \text{Length} + \text{Breadth})$$

Q13. Rectangle with Width Less Than Length

$$\text{Perimeter} = 2 \times (\text{Length} + \text{Breadth})$$

$$\text{Breadth} = \text{Length} - 2$$

$$\text{Area} = \text{Length} \times \text{Breadth}$$

Q14. Rectangular Courtyard

$$\text{Area} = \text{Length} \times \text{Breadth}$$

New Area:

$$(\text{Length} + 2) \times \text{Breadth}$$

Increase in area:

$$[(\text{Length} + 2) \times \text{Breadth}] - (\text{Length} \times \text{Breadth})$$

Q15. Rectangle Area Increase

$$\text{Increase in Area} = \text{Increase in Length} \times \text{Width}$$

$$32 = 8 \times \text{Width}$$

Q16. Rectangle with Diagonal Relation

Area = Length $\times$ Breadth

Diagonal = $\sqrt{[(\text{Length})^2 + (\text{Breadth})^2]}$

Relation:

Diagonal + Length = 3 $\times$ Breadth

Q17. Rectangle with Same Area

Original Area:

Area = Length $\times$ Breadth

Changed Area:

(Length + 15)(Breadth – 10)

Condition:

Length $\times$ Breadth = (Length + 15)(Breadth – 10)

Perimeter:

Perimeter = 2 $\times$ (Length + Breadth)

Q18. Rectangle and Square Equal Area

Area of Square = (Side)²

Area of Rectangle = Length $\times$ Breadth

Breadth = Area / Length

Q19. Square with Poles

Perimeter = Number of Gaps $\times$ Distance

Side = Perimeter / 4

Area = (Side)²

Q20. Square Field Fencing

1 hectare = 10000 square metres

Area of Square = (Side)²

Side = $\sqrt{\text{Area}}$

Perimeter = 4 $\times$ Side

Cost = Perimeter $\times$ Rate

Q21. Square Room and Verandah

Verandah Area = Outer Area – Inner Area

Outer Side = Inner Side + (2 $\times$ Width)

Equation:

(Outer Side)² – (Inner Side)²

Q22. Parallelogram

Area = Base $\times$ Height

Q23. Trapezium

Area = $(1/2) \times (\text{Sum of Parallel Sides}) \times \text{Height}$

where:

- Parallel sides = two parallel sides of trapezium
- Height = perpendicular distance between them

Q24. Grazing Horse

Area Grazed = Area of Quadrant

Area = $(\pi \times \text{Radius}^2) / 4$

where:

- Radius = rope length

Q25. Rectangular Garden with Road

Road Area = Outer Area – Inner Area

Outer Dimensions:

(Length + 10) and (Breadth + 10)

Equation:

$[(\text{Length} + 10)(\text{Breadth} + 10)] - (\text{Length} \times \text{Breadth})$

Q26. Rectangular Field with Path

Path Area = Outer Area – Inner Area

Equation:

$$(\text{Length} \times \text{Breadth}) - [(\text{Length} - 2 \times \text{Width})(\text{Breadth} - 2 \times \text{Width})]$$

Q27. Square Garden with Pavements

Outer Pavement Area:

$$(\text{Outer Side})^2 - (\text{Original Side})^2$$

Inner Pavement Area:

$$(\text{Original Side})^2 - (\text{Inner Side})^2$$

Total Pavement Area:

$$\text{Outer Pavement Area} + \text{Inner Pavement Area}$$

Q28. Rhombus

$$\text{Area} = (1/2) \times \text{Diagonal1} \times \text{Diagonal2}$$

$$\text{Side} = \sqrt{[(\text{Diagonal1} / 2)^2 + (\text{Diagonal2} / 2)^2]}$$

where:

- Diagonal1 and Diagonal2 are diagonals of rhombus